

# Simulation study

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This document aims to reproduce the simulation study ‘Introduction to statistical simulations in health research’ by (Boulesteix et al. 2020).

```
#load packages
#| echo: false
#| output: false
library(Hmisc)
```

Attaching package: 'Hmisc'

The following objects are masked from 'package:base':

```
format.pval, units
```

```
library(mice)
```

Attaching package: 'mice'

The following object is masked from 'package:stats':

```
filter
```

The following objects are masked from 'package:base':

```
cbind, rbind
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.0      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks mice::filter(), stats::filter()
```

```
x dplyr::lag() masks stats::lag()
```

```
x dplyr::src() masks Hmisc::src()
```

```
x dplyr::summarize() masks Hmisc::summarize()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

## Data prep

The different data sets are being loaded:

```
d1 <- sasxport.get("DEMO_I.xpt")
d2 <- sasxport.get("BPX_I.xpt")
d3 <- sasxport.get("BMX_I.xpt")
d4 <- sasxport.get("GHB_I.xpt")
d5 <- sasxport.get("TCHOL_I.xpt")
d1.t <- subset(d1,select=c("seqn","riagendr","ridageyr"))
d2.t <- subset(d2,select=c("seqn","bpxsy1"))
d3.t <- subset(d3,select=c("seqn","bmxbmi"))
d4.t <- subset(d4,select=c("seqn","lbxgh"))
d5.t <- subset(d5,select=c("seqn","lbdtcsl"))
d1.t <- subset(d1,select=c("seqn","riagendr","ridageyr"))
d2.t <- subset(d2,select=c("seqn","bpxsy1"))
d3.t <- subset(d3,select=c("seqn","bmxbmi"))
d4.t <- subset(d4,select=c("seqn","lbxgh"))
d5.t <- subset(d5,select=c("seqn","lbdtcsl"))
d <- merge(d1.t,d2.t)
d <- merge(d,d3.t)
d <- merge(d,d4.t)
d <- merge(d,d5.t)
```

New variables are added to the dataset representing the same variables but a more easily comprehensive names:

```
d$age <- d$ridageyr
d$sex <- d$riagendr
d$bp <- d$bpxsy1
d$bmi <- d$bmx bmi
d$HbA1C <- d$lbxgh
d$chol <- d$lbdtcsi
```

Minors are set to NA and only complete cases are being kept:

```
d$age[d$age<18] <- NA
dc <- cc(subset(d,select=c("age","sex","bmi","HbA1C","bp")))
```

## Analysis

### Assessing the impact of Glycohemoglobin level on systolic blood pressure assuming no measurement errors

A linear regression is run with systolic blood pressure *bp* as the dependent variable and Glycohemoglobin level *HbA1C* as the main predictor, adjusted for age and gender.

```
summary(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

Call:

```
lm(formula = bp ~ HbA1C + age + as.factor(sex), data = dc)
```

Residuals:

```
Systolic: Blood pres (1st rdg) mm Hg
      Min       1Q   Median       3Q      Max
-49.887 -10.509  -1.378   8.491 107.583
```

Coefficients:

|                 | Estimate | Std. Error | t value | Pr(> t )     |
|-----------------|----------|------------|---------|--------------|
| (Intercept)     | 98.75149 | 1.21418    | 81.332  | < 2e-16 ***  |
| HbA1C           | 1.12638  | 0.20291    | 5.551   | 2.98e-08 *** |
| age             | 0.44486  | 0.01284    | 34.648  | < 2e-16 ***  |
| as.factor(sex)2 | -3.24792 | 0.45164    | -7.191  | 7.34e-13 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.1 on 5088 degrees of freedom

Multiple R-squared: 0.2305, Adjusted R-squared: 0.23

F-statistic: 508 on 3 and 5088 DF, p-value: < 2.2e-16

The coefficient for Glycohemoglobin (HbA1C) on systolic blood pressure, controlling for age and sex, is extracted:

```
round(lm(bp ~ HbA1C + age + as.factor(sex), data=dc)$coef[2],2)
```

HbA1C

1.13

The 95% confidence interval of the regression parameters:

```
confint(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

|                 | 2.5 %      | 97.5 %      |
|-----------------|------------|-------------|
| (Intercept)     | 96.3711755 | 101.1317982 |
| HbA1C           | 0.7285836  | 1.5241825   |
| age             | 0.4196932  | 0.4700355   |
| as.factor(sex)2 | -4.1333281 | -2.3625106  |

The second regression with the Body Mass Index of the respondent *bmi* as a confounder.:

```
summary(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

Call:

```
lm(formula = bp ~ HbA1C + bmi + age + as.factor(sex), data = dc)
```

Residuals:

Systolic: Blood pres (1st rdg) mm Hg

| Min     | 1Q      | Median | 3Q    | Max     |
|---------|---------|--------|-------|---------|
| -51.068 | -10.251 | -1.504 | 8.264 | 107.410 |

Coefficients:

| Estimate | Std. Error | t value | Pr(> t ) |
|----------|------------|---------|----------|
|----------|------------|---------|----------|

```

(Intercept)      92.65583      1.39320  66.506 < 2e-16 ***
HbA1C            0.75177      0.20596   3.650 0.000265 ***
bmi             0.28632      0.03282   8.724 < 2e-16 ***
age             0.44586      0.01275  34.979 < 2e-16 ***
as.factor(sex)2 -3.63115      0.45049  -8.060 9.4e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 15.98 on 5087 degrees of freedom
Multiple R-squared:  0.2418,    Adjusted R-squared:  0.2412
F-statistic: 405.7 on 4 and 5087 DF,  p-value: < 2.2e-16

```

The effect of Glycohemoglobin on systolic blood pressure, adjusted for age, sex, and BMI, is reported as:

```
round(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc)$coef[2],2)
```

```

HbA1C
0.75

```

With the CI of:

```
confint(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

```

                2.5 %      97.5 %
(Intercept)    89.9245592 95.3871089
HbA1C          0.3479966  1.1555348
bmi            0.2219815  0.3506673
age            0.4208695  0.4708464
as.factor(sex)2 -4.5143014 -2.7479929

```

### Assessing the impact of Glycohemoglobin on systolic blood pressure in the presence of measurement errors

The goal is to assess how measurement errors in both the exposure and the confounder affect the regression results.

The reference value for the impact of Glycohemoglobin on systolic blood pressure, without measurement error:

```
ref <- lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc)$coef[2]
```

*HbA1C.me* and *bmi.me* are then created in a simulation process of 1000 datasets with variance of the measurement errors between 0 and 0.5 of their original variances:

```
n.sim <- 1e3
perc.me.exp <- seq(0,.5,.1)
perc.me.conf<- seq(0,.5,.1)
scenarios <- expand.grid(perc.me.exp,perc.me.conf)
```

The reference variances in the original exposure and confounder variables are stored in “var.exp” and “var.conf”:

```
var.exp <- var(dc$HbA1C)
var.conf <- var(dc$bmi)
```

The original sample size:

```
n <- dim(dc)[1]
```

The empty matrix “beta.hat” is created to store the regression coefficients of *HbA1C.me*.

```
beta.hat <- matrix(ncol=dim(scenarios)[1], nrow=n.sim)
```

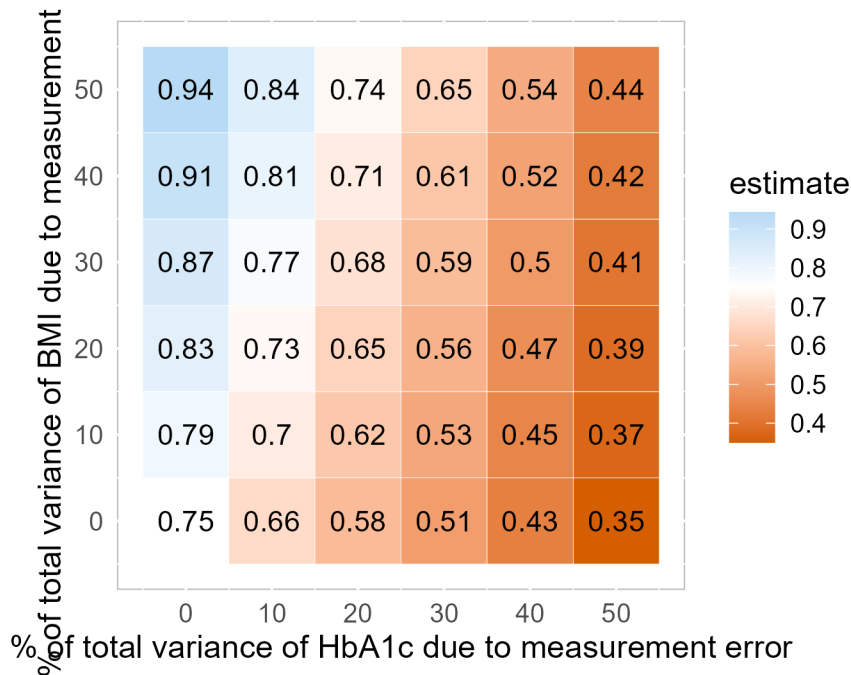
The simulation process involves nested loops for each of the 1000 simulations and 36 scenarios, where *HbA1C.me* and *bmi.me* are generated by adding measurement errors randomly sampled from normal distributions:

```
for (k in 1:n.sim){
  print(k)
  set.seed(k)
  for (i in 1:dim(scenarios)[1]){
    var.me.exp <- var.exp*scenarios[i,1]/(1-scenarios[i,1])
    var.me.conf <- var.conf*scenarios[i,2]/(1-scenarios[i,2])
    dc$HbA1C.me <- dc$HbA1C + rnorm(dim(dc)[1], 0, sqrt(var.me.exp) )
    dc$bmi.me <- dc$bmi + rnorm(dim(dc)[1], 0, sqrt(var.me.conf) )
    beta.hat[k,i] <- lm(bp ~ HbA1C.me + age + bmi.me + as.factor(sex), data=dc)$coef[2]
  }}
}
```

## Results

The result of the simulation is visualized using a heatmap showing how the impact of Glycohemoglobin on systolic blood pressure varies according to different levels of residual variance for both the exposure *HbA1C* and confounder *BMI*. The data is stored in the `tot.mat` matrix and plotted using the `ggplot2` package:

The figure matches the figure obtained by Boulesteix, Groenwold, Abrahamowicz, et al. (2020).



Boulesteix, Anne-Laure, Rolf Hh Groenwold, Michal Abrahamowicz, Harald Binder, Matthias Briel, Roman Hornung, Tim P Morris, Jörg Rahnenführer, and Willi Sauerbrei. 2020. "Introduction to Statistical Simulations in Health Research." *BMJ Open* 10 (12): e039921. <https://doi.org/10.1136/bmjopen-2020-039921>.